Similar questions

Five new questions practising the same skills as the exam question above. They are not past exam questions. Marks are as in the original. Answers are at the end.

1 In this question you must show detailed reasoning.

(i) Express $2^{7/2}$ in the form $a\sqrt{b}$, where a is an integer and b is a prime number. **[2]**

(ii) Express $\dfrac{\sqrt{3}}{2-\sqrt{3}}$ in the form $c+d\sqrt{e}$, where c and d are integers and e is a prime number. **[3]**

2 In this question you must show detailed reasoning.

(i) Express $3^{9/2}$ in the form $a\sqrt{b}$, where a is an integer and b is a prime number. **[2]**

(ii) Express $\dfrac{\sqrt{5}}{2-\sqrt{5}}$ in the form $c+d\sqrt{e}$, where c and d are integers and e is a prime number. **[3]**

3 In this question you must show detailed reasoning.

(i) Express $2^{9/2}$ in the form $a\sqrt{b}$, where a is an integer and b is a prime number. **[2]**

(ii) Express $\dfrac{\sqrt{2}}{3-2\sqrt{2}}$ in the form $c+d\sqrt{e}$, where c and d are integers and e is a prime number. **[3]**

4 In this question you must show detailed reasoning.

(i) Express $5^{7/2}$ in the form $a\sqrt{b}$, where a is an integer and b is a prime number. **[2]**

(ii) Express $\dfrac{2}{\sqrt{5}-2}$ in the form $c+d\sqrt{e}$, where c and d are integers and e is a prime number. **[3]**

5 In this question you must show detailed reasoning.

(i) Express $7^{5/2}$ in the form $a\sqrt{b}$, where a is an integer and b is a prime number. **[2]**

(ii) Express $\dfrac{3+\sqrt{2}}{1-\sqrt{2}}$ in the form $c+d\sqrt{e}$, where c and d are integers and e is a prime number. **[3]**

Question 1

(i) $2^{7/2} = 2^3\sqrt{2} = 8\sqrt{2} \quad (a = 8,\ b = 2)$

(ii) $\dfrac{\sqrt{3}}{2-\sqrt{3}} \times \dfrac{2+\sqrt{3}}{2+\sqrt{3}} = \dfrac{2\sqrt{3}+3}{4-3} = 3+2\sqrt{3}$

Question 2

(i) $3^{9/2} = 3^4\sqrt{3} = 81\sqrt{3}$

(ii) $\dfrac{\sqrt{5}}{2-\sqrt{5}} \times \dfrac{2+\sqrt{5}}{2+\sqrt{5}} = \dfrac{2\sqrt{5}+5}{4-5} = -5-2\sqrt{5}$

Question 3

(i) $2^{9/2} = 2^4\sqrt{2} = 16\sqrt{2}$

(ii) $\dfrac{\sqrt{2}}{3-2\sqrt{2}} \times \dfrac{3+2\sqrt{2}}{3+2\sqrt{2}} = \dfrac{3\sqrt{2}+4}{9-8} = 4+3\sqrt{2}$

Question 4

(i) $5^{7/2} = 5^3\sqrt{5} = 125\sqrt{5}$

(ii) $\dfrac{2}{\sqrt{5}-2} \times \dfrac{\sqrt{5}+2}{\sqrt{5}+2} = \dfrac{2\sqrt{5}+4}{5-4} = 4+2\sqrt{5}$

Question 5

(i) $7^{5/2} = 7^2\sqrt{7} = 49\sqrt{7}$

(ii) $\dfrac{(3+\sqrt{2})(1+\sqrt{2})}{(1-\sqrt{2})(1+\sqrt{2})} = \dfrac{5+4\sqrt{2}}{-1} = -5-4\sqrt{2}$